

Image Processing – Fourier Extraction

Finger-Vein PAD using Fourier Magnitude and Phase Features

Andreas Auer Philipp Fuchs Sarah Götz

4th February 2026

- Fourier Introduction
- Task 1
 - Experiments
 - Results
- Task 2
 - Experiments
 - Results
- Final Results & Conclusion

2026-02-04

Image Processing – Fourier Extraction

└ Outline

Hi

OUTLINE

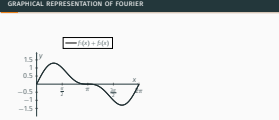
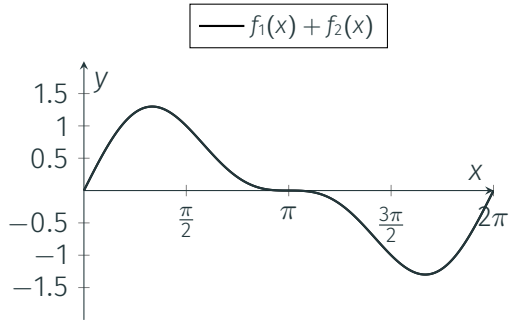
- Fourier Introduction
- Task 1
 - Experiments
 - Results
- Task 2
 - Experiments
 - Results
- Final Results & Conclusion

2026-02-04

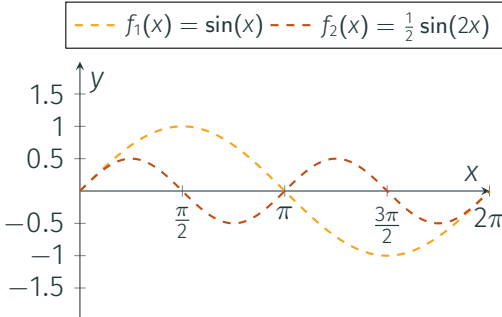
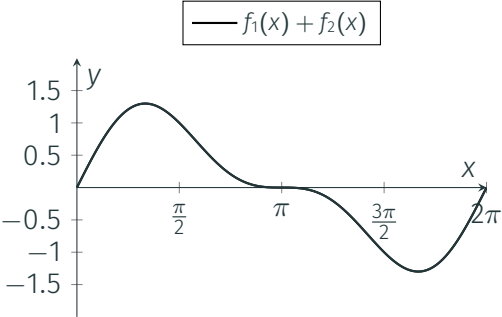
Image Processing – Fourier Extraction
└─ Fourier Introduction

Fourier Introduction

Fourier Introduction



Quick recap



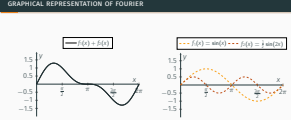
2026-02-04

Image Processing – Fourier Extraction

└ Fourier Introduction

└ Graphical Representation of Fourier

Quick recap



Fourier-Series

$$f(t) = \sum_{k=-\infty}^{\infty} c_k e^{ik\omega_0 t}$$

- k : index of the frequency components
- $\omega_0 = \frac{2\pi}{T}$: fundamental angular frequency vibration
- $c_k \in \mathbb{C}$: complex fourier coefficients

- complex fourier coefficients
- angular frequency vibration

$$f(t) = \sum_{k=-\infty}^{\infty} c_k e^{ik\omega_0 t}$$

- k : index of the frequency components
- $\omega_0 = \frac{2\pi}{T}$: fundamental angular frequency vibration
- $c_k \in \mathbb{C}$: complex fourier coefficients

$$c_k = \text{Re}(c_k) + i \text{Im}(c_k)$$

$$|c_k| = \sqrt{\text{Re}(c_k)^2 + \text{Im}(c_k)^2} \quad \text{(Magnitude)}$$

$$\varphi_k = \arg(c_k) \iff c_k = |c_k| e^{i\varphi_k} \quad \text{(Phase)}$$

2026-02-04

Image Processing – Fourier Extraction
└ Fourier Introduction

└ Magnitude and Phase of Fourier Coefficients

- magnitude: strength of a frequency
- phase: positional information
- We experiment with both of them and check which results we get

MAGNITUDE AND PHASE OF FOURIER COEFFICIENTS

$$c_k = \text{Re}(c_k) + i \text{Im}(c_k)$$
$$|c_k| = \sqrt{\text{Re}(c_k)^2 + \text{Im}(c_k)^2} \quad \text{(Magnitude)}$$
$$\varphi_k = \arg(c_k) \iff c_k = |c_k| e^{i\varphi_k} \quad \text{(Phase)}$$

$$c_k = \text{Re}(c_k) + i \text{Im}(c_k)$$

$$|c_k| = \sqrt{\text{Re}(c_k)^2 + \text{Im}(c_k)^2} \quad (\text{Magnitude})$$

$$\varphi_k = \arg(c_k) \iff c_k = |c_k| e^{i\varphi_k} \quad (\text{Phase})$$

Magnitude $|c_k|$

→ Strength of the frequency $k \cdot \omega_0$.

Phase φ_k

→ Gives a “position” relative to the x-Axis.

2026-02-04

Image Processing – Fourier Extraction

└ Fourier Introduction

└ Magnitude and Phase of Fourier Coefficients

- magnitude: strength of a frequency
- phase: positional information
- We experiment with both of them and check which results we get

MAGNITUDE AND PHASE OF FOURIER COEFFICIENTS

$$c_k = \text{Re}(c_k) + i \text{Im}(c_k)$$
$$|c_k| = \sqrt{\text{Re}(c_k)^2 + \text{Im}(c_k)^2} \quad (\text{Magnitude})$$
$$\varphi_k = \arg(c_k) \iff c_k = |c_k| e^{i\varphi_k} \quad (\text{Phase})$$

Magnitude $|c_k|$
→ Strength of the frequency $k \cdot \omega_0$.

Phase φ_k
→ Gives a “position” relative to the x-Axis.

- **Fourier** feature extraction.
→ 2D Fast-Fourier-Transformation (**FFT**)
- FFT returns **N** radial bands → N -dimensional feature vector $\in \mathbb{R}^N$ for:
 1. **Magnitude features:**
 - Power spectrum, summed over each radial band.
 2. **Phase features:**
 - Circular mean of phase values per band.

(u, v) : frequency coordinates in the 2D Fourier domain,

$$B_i := \{(u, v) \mid r_i \leq \sqrt{u^2 + v^2} < r_{i+1}\} \text{ (radial band } i),$$

$$i = 1, \dots, N.$$

$$v_{\text{mag}}[i] = \sum_{(u,v) \in B_i} |\text{FFT}(u, v)|$$

$$v_{\text{phase}}[i] = \sum_{(u,v) \in B_i} \arg(\text{FFT}(u, v))$$

- L2-normalization:

- Removes scale differences by projecting onto the **unit sphere**.
⇒ Improves comparability.
- Focus on **patterns** rather than **intensity/brightness** variations.
- **Always** used in our experiments.

- projection onto unit sphere
- focus on patterns
- not on intensity/brightness

- Removes scale differences by projecting onto the **unit sphere**.
⇒ Improves comparability.
- Focus on **patterns** rather than **intensity/brightness** variations.
- **Always** used in our experiments.

- **L2-normalization:**
 - Removes scale differences by projecting onto the **unit sphere**.
 $\implies$ Improves comparability.
 - Focus on **patterns** rather than **intensity/brightness** variations.
 - **Always** used in our experiments.
- **Log-compression (magnitude only):**
 - **Compresses** (“shrinks”) large magnitude spikes
 $\implies$ prevents them from **dominating** the representation.
 - **Spreads** out small magnitude values.
 $\implies$ Subtle details become more visible and **more influential** for learning.
 - Stabilizes features: multiplicative changes $\rightarrow$ **additive** shifts.
 - **Experiment-based** use.

Experiments Task 1

2026-02-04

Image Processing – Fourier Extraction
└ Experiments Task 1

Experiments Task 1

- Let's get to our Experiments for Task 1

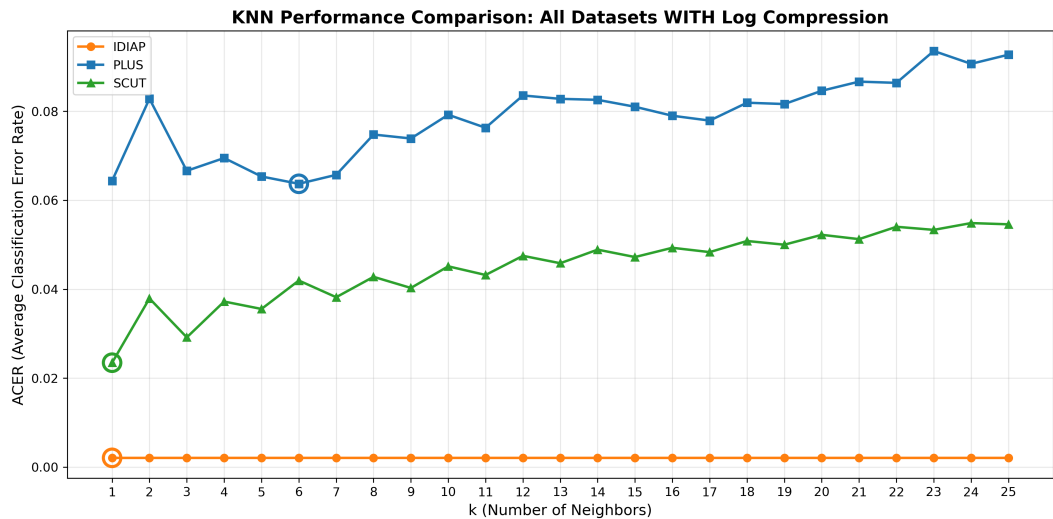
Evaluation criterion: ACER of the baseline configuration

- Optimize k in KNN
 - Search over $k \in [1, 25]$
- Magnitude vs. Phase
 - Evaluate magnitude-only, phase-only, and combined features
- Inter-Class Variation
 - Real vs. Spoof
- Intra-Class Variation
 - Variation within Real and within Spoof samples
- Effect of Log-Compression
 - Compare magnitude features w/ vs. w/o log-compression

- First of all: Evaluation criterion: ACER of baseline config
- First exp: optimizing k
- As we heard before: experiment with magnitude and phase – why?, mag-only, phase-only, combinations
- To explain results: inter-class and intra-class variation
- Lastly we show the effect of log-compression (and mentioned before)

- Optimize k in KNN
 - Search over $k \in [1, 25]$
- Magnitude vs. Phase
 - Evaluate magnitude-only, phase-only, and combined features
- Inter-Class Variation
 - Real vs. Spoof
- Intra-Class Variation
 - Variation within Real and within Spoof samples
- Effect of Log-Compression
 - Compare magnitude features w/ vs. w/o log-compression

EVALUATE BEST k FOR KNN



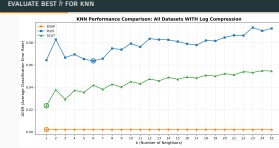
2026-02-04

Image Processing – Fourier Extraction

└ Experiments Task 1

└ Evaluate best k for KNN

└ Evaluate best k for KNN



- x-axis: k
- y-axis: ACER
- all datasets: $k=1$ good
- IDIAP (orange), SCUT (green) $k=1$ best
- PLUS (blue): $k=6$ slightly better

- *Best k*: PLUS: 6, IDIAP: 1, SCUT: 1

2026-02-04

- Image Processing – Fourier Extraction
 - └ Experiments Task 1
 - └ Evaluate best *k* for KNN
 - └ Results Baseline

- Therefore first result/setting: best k's

RESULTS BASELINE

- *Best k*: PLUS: 6, IDIAP: 1, SCUT: 1

Pearson correlation of magnitude and phase feature vectors:

- PLUS: ≈ 0.048
- IDIAP: ≈ -0.010
- SCUT: ≈ 0.016

⇒ Features are almost **uncorrelated** – a **combination** may provide complementary information.

⇒ Let's try it.

2026-02-04

Image Processing – Fourier Extraction

└ Experiments Task 1

└ Comparing combinations of magnitude and phase

└ Correlation of Magnitude and Phase

- Now we get to our magnitude vs phase comparison/experiments
- First: Pearson correlation => almost uncorrelated => let's try combining them (may provide complementary information)

Pearson correlation of magnitude and phase feature vectors:

- PLUS: ≈ 0.048
- IDIAP: ≈ -0.010
- SCUT: ≈ 0.016

⇒ Features are almost **uncorrelated** – a **combination** may provide complementary information.

⇒ Let's try it.

$$M = [1, \dots, 10, 11, \dots, 20, 21, \dots, 30]$$

$M_1 \quad M_2 \quad M_3$

$$P = [1, \dots, 10, 11, \dots, 20, 21, \dots, 30]$$

$P_1 \quad P_2 \quad P_3$

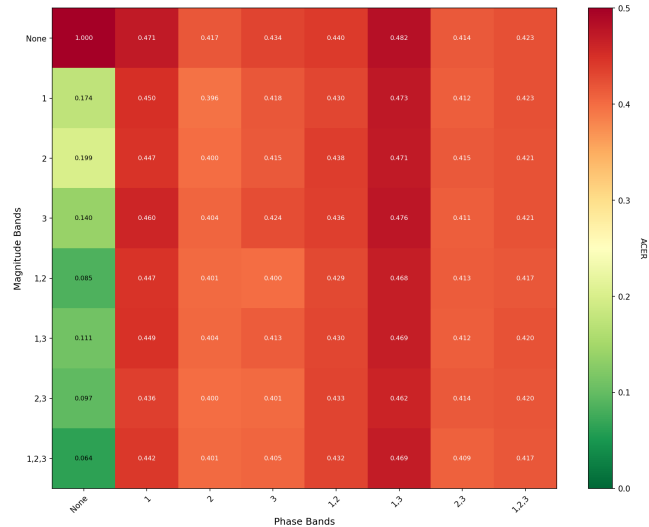
$$\mathcal{P}(M) = \{\emptyset, \{M_1\}, \{M_2\}, \{M_3\}, \{M_1, M_2\}, \{M_1, M_3\}, \{M_2, M_3\}, \{M_1, M_2, M_3\}\}$$

$$\mathcal{P}(P) = \{\emptyset, \{P_1\}, \{P_2\}, \{P_3\}, \{P_1, P_2\}, \{P_1, P_3\}, \{P_2, P_3\}, \{P_1, P_2, P_3\}\}$$

$$\mathcal{P}(M) \times \mathcal{P}(P) = \{\emptyset, \{\emptyset, P_1\}, \{\emptyset, P_2\}, \dots, \{M_1, M_2, M_3, P_1, P_2, P_3\}\}$$

- First I need to explain the setup for how we combined these features.
- Didn't just combined the whole 30 dimensional feature vectors
- Instead: split into thirds (see e.g. mag vector...)
- constructed powersets
- built cross product of the two powersets to get all the possible combinations

COMBINING DIFFERENT THIRDS OF MAGNITUDE AND PHASE – PLUS DATASET



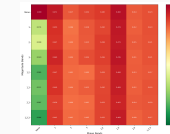
2026-02-04

Image Processing – Fourier Extraction

Experiments Task 1

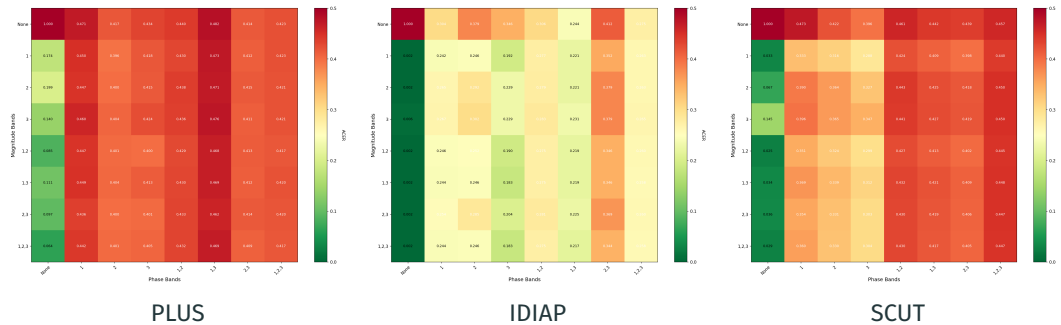
Comparing combinations of magnitude and phase

Combining Different Thirds of Magnitude and Phase – PLUS Dataset



- heatmap with these 64 combinations; here plus dataset
- x-axis: phase powerset
- y-axis: magnitude powerset
- empty set being none here
- the greener the better the results (lower ACER); red is bad obv
- key observation: combination with no phase bands and some mag bands (left strip) clearly perform the best
- more precisely all thirds and no phase bands is the winner

COMBINING DIFFERENT THIRDS OF MAGNITUDE AND PHASE – ALL DATASETS



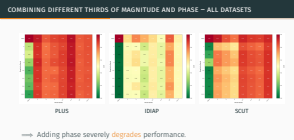
⇒ Adding phase severely **degrades** performance.

2026-02-04

Image Processing – Fourier Extraction

Experiments Task 1

- Comparing combinations of magnitude and phase
- Combining Different Thirds of Magnitude and Phase – all Datasets



- same pattern on the other datasets too
- adding phase just severely degrades performance
- sometimes a little more, sometimes a little less
- but always the same pattern and clearly we can say...
- ... [NEXT] magnitude only is best for performance – phase worsens the results

- Best k : PLUS: 6, IDIAP: 1, SCUT: 1
- **Magnitude only** is best for performance – Phase worsens the results.

2026-02-04

Image Processing – Fourier Extraction

└ Experiments Task 1

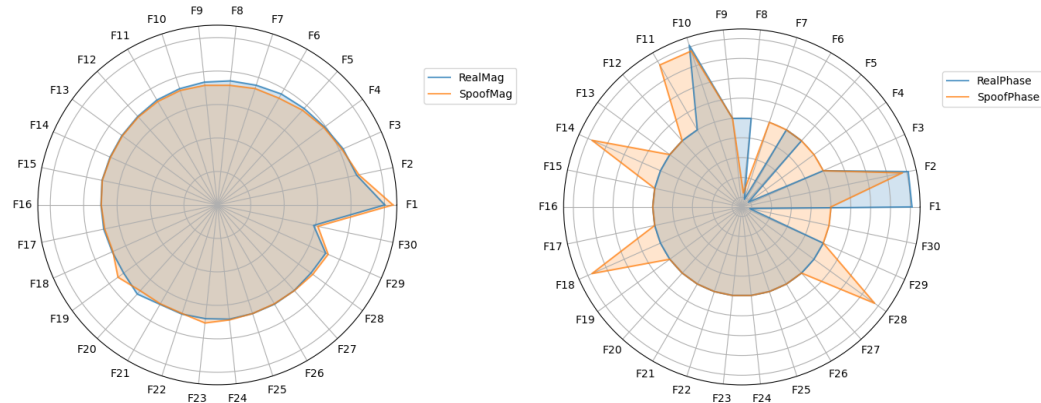
└ Comparing combinations of magnitude and phase

└ Results Baseline

- magnitude only is best for performance
- phase worsens the results
- now lets try to find an explanations why these results occur

- Best k : PLUS: 6, IDIAP: 1, SCUT: 1
- **Magnitude only** is best for performance – Phase worsens the results.

INTER-CLASS VARIATION: REAL VS. SPOOF



⇒ **Magnitude** features: small differences → **low** Inter-Class Variation.

⇒ **Phase** features: large variations → **high** Inter-Class Variation.

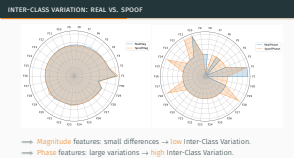
2026-02-04

Image Processing – Fourier Extraction

└ Experiments Task 1

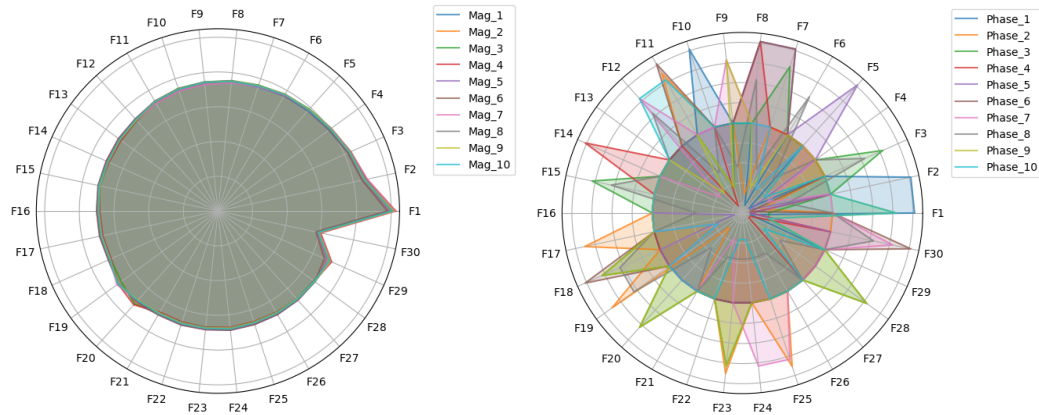
└ Comparing combinations of magnitude and phase

└ Inter-Class Variation: Real vs. Spoof



- inter-class variation => how different are reals and real spoofs between each other
- spider plots with the 30 dimension / features of the feature vector displayed independently
- left: mag of real and real spoofs; right: phase of real and real spoofs
- mag features: small (but noticeably) diff; low inter...; phase features: large var; high inter...
- From this one could think: phase has greater variations => more differences to learn for the model => easier to distinguish

INTRA-CLASS VARIATION: REAL



- ⇒ **Magnitude** features: very consistent → **low** Intra-Class Variation.
- ⇒ **Phase** features: highly irregular/random → **high** Intra-Class Variation.

2026-02-04

Image Processing – Fourier Extraction

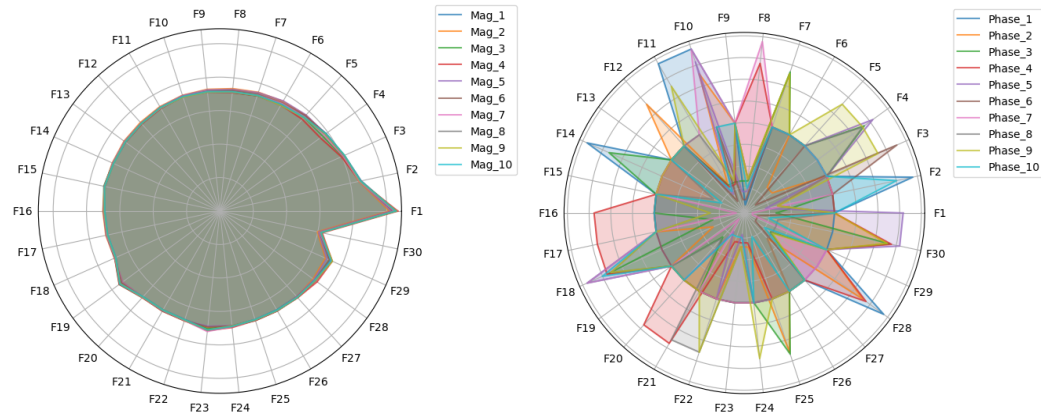
Experiments Task 1

- Comparing combinations of magnitude and phase
- Intra-Class Variation: Real



- intra-class variation: how different are the magnitudes and phases in one class
- here: Reals
- left magnitude of 10 reals; right phase of 10 reals
- while mag is very consistent; low intra
- phase is highly irregular; basically random (christmas tree from above)
- phase intra-class var. too high for the model to learn anything

INTRA-CLASS VARIATION: SPOOF

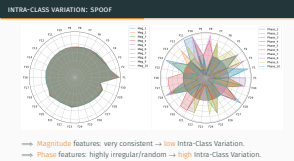


- ⇒ **Magnitude** features: very consistent → **low** Intra-Class Variation.
- ⇒ **Phase** features: highly irregular/random → **high** Intra-Class Variation.

2026-02-04

Image Processing – Fourier Extraction

- └ Experiments Task 1
 - └ Comparing combinations of magnitude and phase
 - └ Intra-Class Variation: Spoof



- same pattern for the reals Spoofs

- Best k : PLUS: 6, IDIAP: 1, SCUT: 1
- **Magnitude only** is best for performance – Phase worsens the results.
 - **Magnitude** features:
consistent within each class + small but stable inter-class differences
⇒ good discriminative power.
 - **Phase** features:
large intra-class variation/randomness
⇒ differences between real and spoof not reliably exploitable.

2026-02-04

Image Processing – Fourier Extraction

└ Experiments Task 1

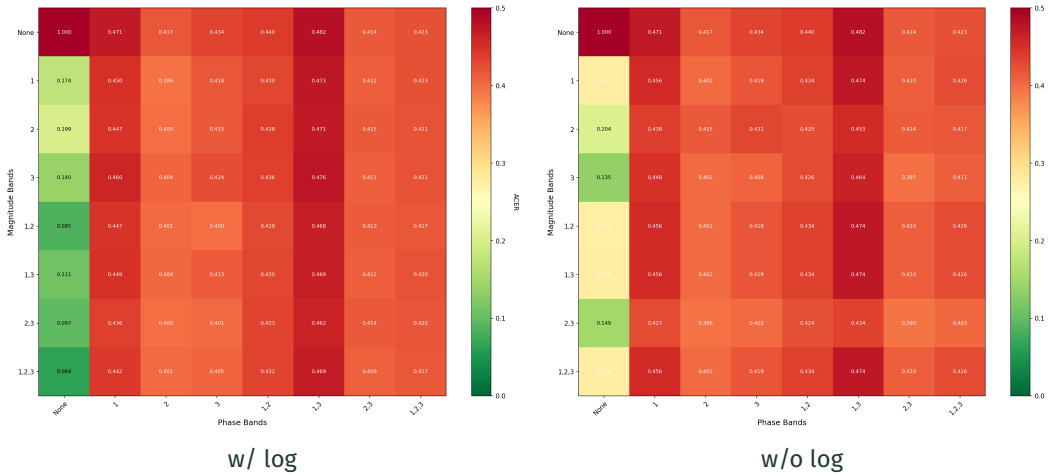
└ Comparing combinations of magnitude and phase

└ Results Baseline

- our conclusion / explanation for magnitude only being the best
- mag: good discriminative power through the consistency within each class and the small but stable inter-class differences
- phase: because of the large intra-class variation / randomness, diff between real and spoof not reliably exploitable

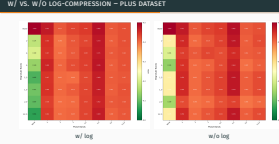
- Best k : PLUS: 6, IDIAP: 1, SCUT: 1
- **Magnitude only** is best for performance – Phase worsens the results.
 - **Magnitude** features:
consistent within each class + small but stable inter-class differences
⇒ good discriminative power.
 - **Phase** features:
large intra-class variation/randomness
⇒ differences between real and spoof not reliably exploitable.

w/ vs. w/o LOG-COMPRESSION – PLUS DATASET



2026-02-04

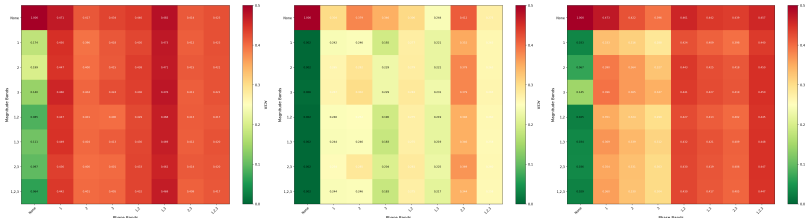
Image Processing – Fourier Extraction
└ Experiments Task 1
└ Comparing combinations of magnitude and phase
└ w/ vs. w/o Log-Compression – PLUS Dataset



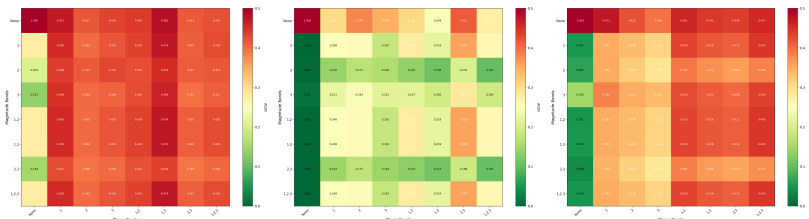
- last exp: impact of log-compression
- left: with log-compression on mag features; right: without log-compression
- clear results: combinations with phase-bands roughly the same
- left strip with magnitude-only worsens when using no log compression

w/ vs. w/o LOG-COMPRESSION – ALL DATASETS

w/ log



w/o log



⇒ Log-Compression improves results (magnitude-only).

2026-02-04

Image Processing – Fourier Extraction

- └ Experiments Task 1
 - └ Comparing combinations of magnitude and phase
 - └ w/ vs. w/o Log-Compression – all Datasets



- again: roughly same pattern for all datasets
- log-compressions improves results in all dataset on the magnitude only strip
- when phase is added there are now some better results without log, but still no way near the good results of magnitude only, so we can ignore that

- **Best k :** PLUS: 6, IDIAP: 1, SCUT: 1
- **Magnitude only** is best for performance – Phase worsens the results.
 - **Magnitude** features:
consistent within each class + small but stable inter-class differences
⇒ good discriminative power.
 - **Phase** features:
large intra-class variation/randomness
⇒ differences between real and spoof not reliably exploitable.
- **Log-Compression** improves results in all datasets.

2026-02-04

Image Processing – Fourier Extraction

└ Experiments Task 1

└ Comparing combinations of magnitude and phase

└ Results Baseline

- adding this last result to our list, gives us the “optimal” settings

- **Best k :** PLUS: 6, IDIAP: 1, SCUT: 1
- **Magnitude only** is best for performance – Phase worsens the results.
 - **Magnitude** features:
consistent within each class + small but stable inter-class differences
⇒ good discriminative power.
 - **Phase** features:
large intra-class variation/randomness
⇒ differences between real and spoof not reliably exploitable.
- **Log-Compression** improves results in all datasets.

2026-02-04

Image Processing – Fourier Extraction
└ Results Task 1

Results Task 1

Results Task 1

k: PLUS: 6, IDIAP: 1, SCUT: 1; Magnitude-only; Log-Compression on; Normalization on

Dataset	Baseline (%)			Step 1 (%)			Step 2 (%)		
	APCER	BPCER	ACER	APCER	BPCER	ACER	APCER	BPCER	ACER
IDIAP	0.00	0.42	0.21	0.00	0.42	0.21	0.00	0.00	0.00
PLUS	6.12	6.33	6.23	9.09	6.67	7.88	10.51	13.67	12.09
SCUT	4.31	0.44	2.38	4.86	0.72	2.79	6.69	0.97	3.83

- ⇒ IDIAP unexpectedly improves,
- ⇒ PLUS degrades notably,
- ⇒ SCUT degrades slightly.

2026-02-04

Image Processing – Fourier Extraction
└ Results Task 1

└ Results Task 1 with our best settings

RESULTS TASK 1 WITH OUR BEST SETTINGS

k: PLUS: 6, IDIAP: 1, SCUT: 1; Magnitude-only; Log-Compression on; Normalization on

Dataset	Baseline (%)			Step 1 (%)			Step 2 (%)		
	APCER	BPCER	ACER	APCER	BPCER	ACER	APCER	BPCER	ACER
IDIAP	0.00	0.42	0.21	0.00	0.42	0.21	0.00	0.00	0.00
PLUS	6.12	6.33	6.23	9.09	6.67	7.88	10.51	13.67	12.09
SCUT	4.31	0.44	2.38	4.86	0.72	2.79	6.69	0.97	3.83

⇒ IDIAP unexpectedly improves,
⇒ PLUS degrades notably,
⇒ SCUT degrades slightly.

- table: dataset; baseline, step 1 and 2
- for better visualization, we added a bit of color: the darker the orange the better the result
- IDIAP: very low all the time; it even improves unexpectedly (0 % at step 2)
- PLUS: degrades notably; factor 2 between baseline and step 2
- SCUT: decgrades slightly

2026-02-04

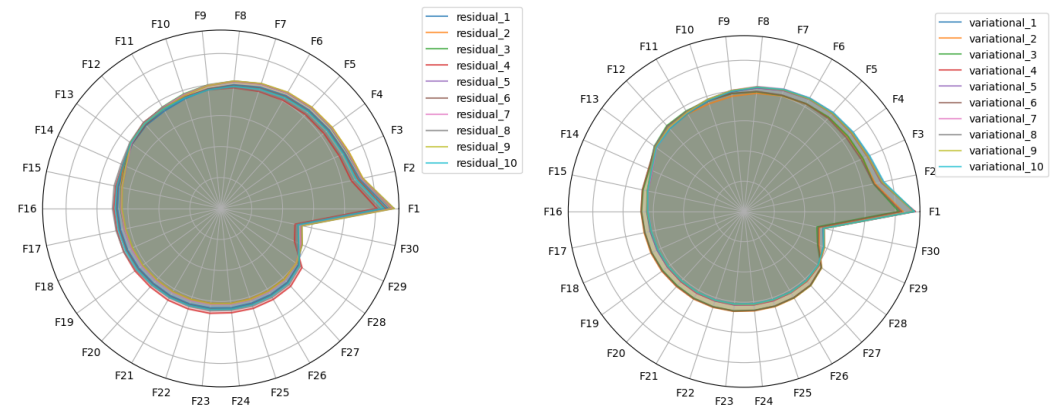
Image Processing – Fourier Extraction
└ Experiments Task 2

Experiments Task 2

Experiments Task 2

Evaluation focus: Behaviour of synthetic spoof data

- **Intra-Class** Variation:
 - Synthetic Spoof (Residual)
 - Synthetic Spoof (Variational)
- **Inter-Class** Variation:
 - Real vs. Synthetic Spoof
 - Spoof vs. Synthetic Spoof
- Comparison of Different **Thirds**
 - Are any feature bands better than others?
 - Explaining the results?



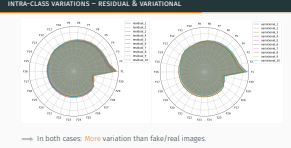
⇒ In both cases: **More** variation than fake/real images.

2026-02-04

Image Processing – Fourier Extraction
└ Experiments Task 2

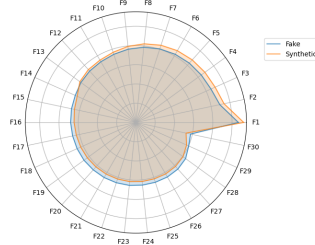
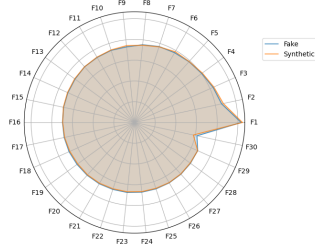
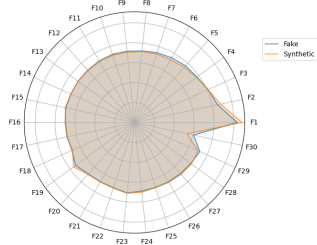
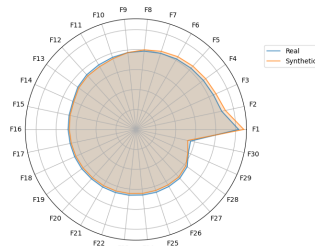
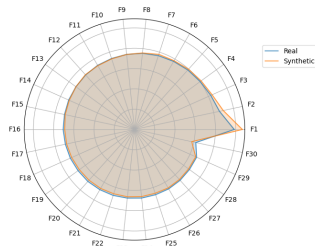
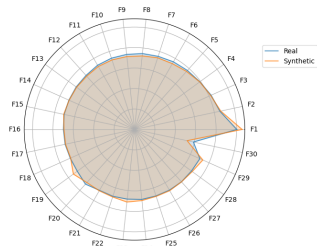
└ Intra-Class Variations – residual & variational

- more variation than fake/real image
- residual and variational looks alike



⇒ In both cases: **More** variation than fake/real images.

INTER-CLASS VARIATIONS – IDIAP & PLUS & SCUT



⇒ IDIAP & PLUS: Small difference to Spoofs – recognizable difference to Reals.

25/36

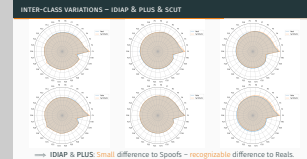
2026-02-04

Image Processing – Fourier Extraction

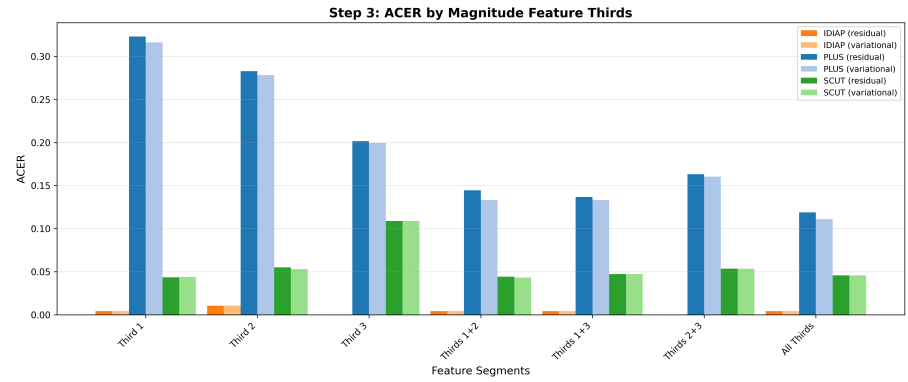
Experiments Task 2

Inter-Class Variations – IDIAP & PLUS & SCUT

- idiap: nice difference
- plus: subtle differences, more like fake
- scut: more like real
- may result in higher APCER - false positive
- look at thirds again



COMPARISON OF DIFFERENT THIRDS – STEP 3



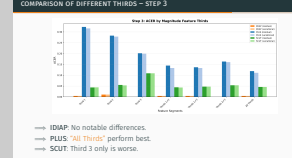
- ⇒ IDIAP: No notable differences.
- ⇒ PLUS: “All Thirds” perform best.
- ⇒ SCUT: Third 3 only is worse.

2026-02-04

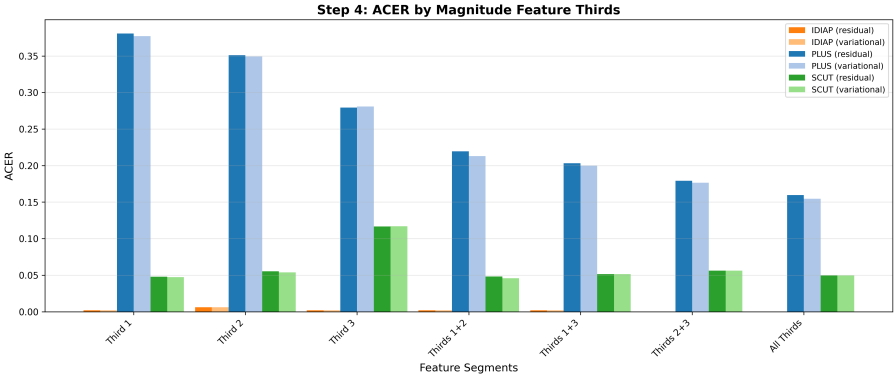
Image Processing – Fourier Extraction └ Experiments Task 2

└ Comparison of different thirds – Step 3

- powerset of magnitude features
- no phase
- look at thirds
- which combination is best
- we use best combination for training



COMPARISON OF DIFFERENT THIRDS – STEP 4



- ⇒ IDIAP: No notable differences
- ⇒ PLUS: “All Thirds” perform best
- ⇒ SCUT: Third 3 only is worse

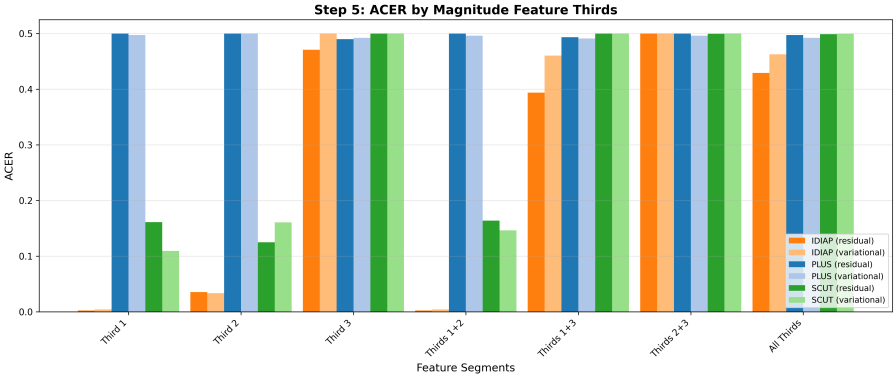
2026-02-04

Image Processing – Fourier Extraction
└ Experiments Task 2

└ Comparison of different thirds – Step 4



COMPARISON OF DIFFERENT THIRDS – STEP 5



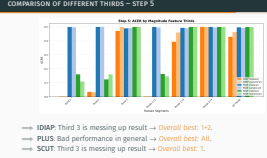
- ⇒ IDIAP: Third 3 is messing up result → Overall best: 1+2.
- ⇒ PLUS: Bad performance in general → Overall best: All.
- ⇒ SCUT: Third 3 is messing up result → Overall best: 1.

2026-02-04

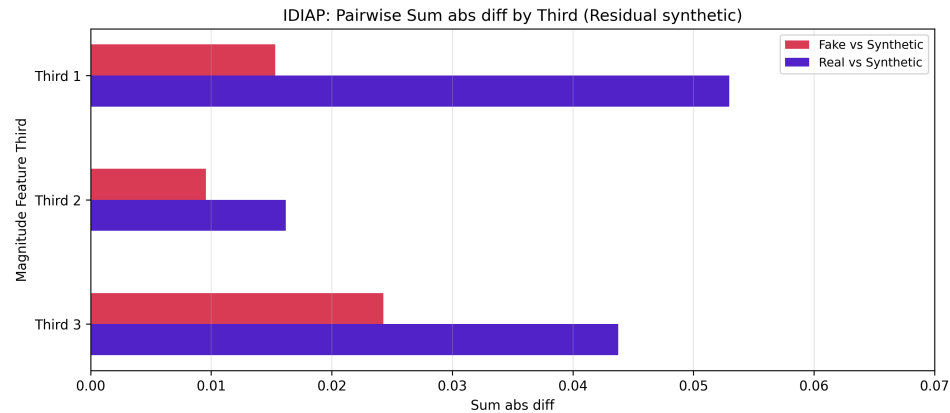
Image Processing – Fourier Extraction └ Experiments Task 2

└ Comparison of different thirds – Step 5

- overall performance!!



EXPLANATION – IDIAP – RESIDUAL



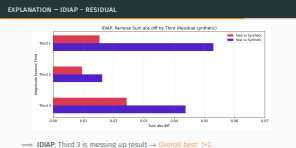
⇒ IDIAP: Third 3 is messing up result → *Overall best: 1+2.*

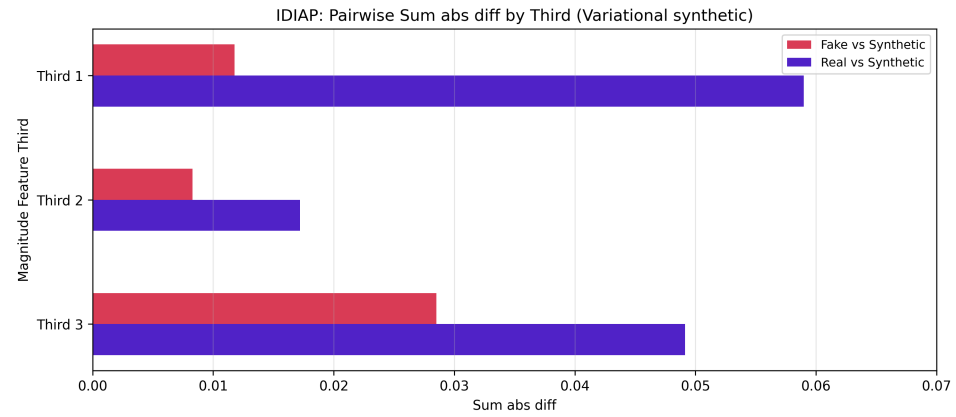
2026-02-04

Image Processing – Fourier Extraction
└ Experiments Task 2

└ Explanation – IDIAP - residual

- F vs S SMALL
- R vs S LARGE
- bar length diff important



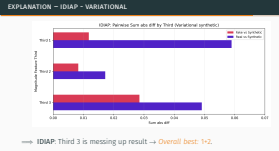


⇒ IDIAP: Third 3 is messing up result → *Overall best: 1+2.*

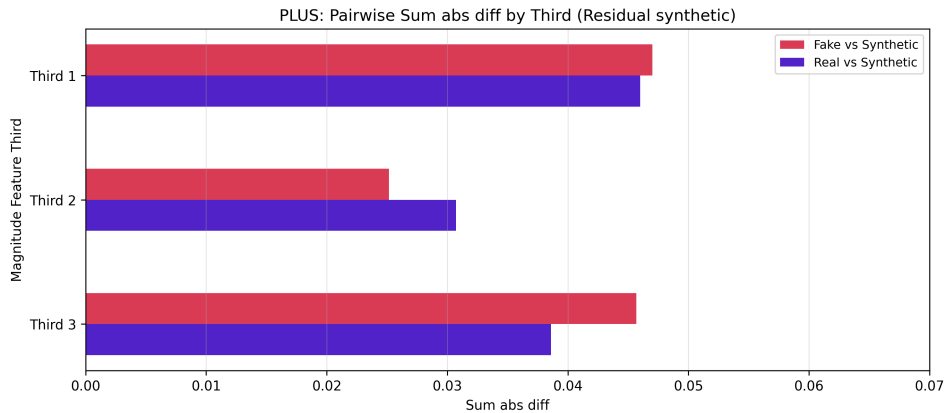
2026-02-04

Image Processing – Fourier Extraction
└ Experiments Task 2

└ Explanation – IDIAP - variational



EXPLANATION – PLUS – RESIDUAL

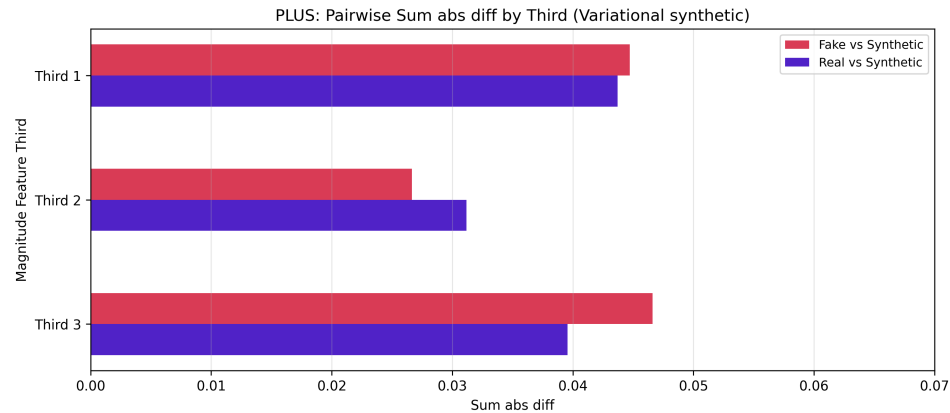


⇒ **PLUS:** Bad performance in general → *Overall best: All.*

2026-02-04
Image Processing – Fourier Extraction
└ Experiments Task 2

└ Explanation – PLUS - residual



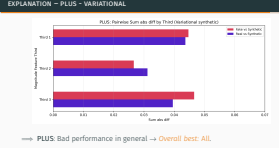


⇒ **PLUS:** Bad performance in general → *Overall best: All.*

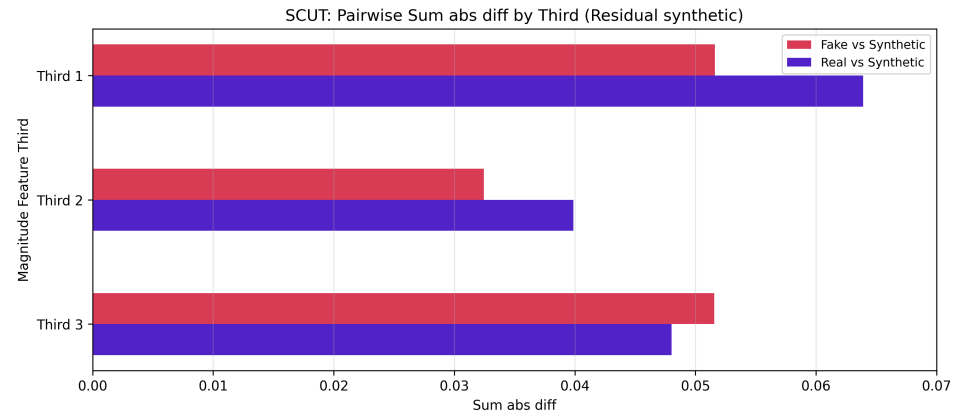
2026-02-04

Image Processing – Fourier Extraction
└ Experiments Task 2

└ Explanation – PLUS - variational



EXPLANATION – SCUT - RESIDUAL

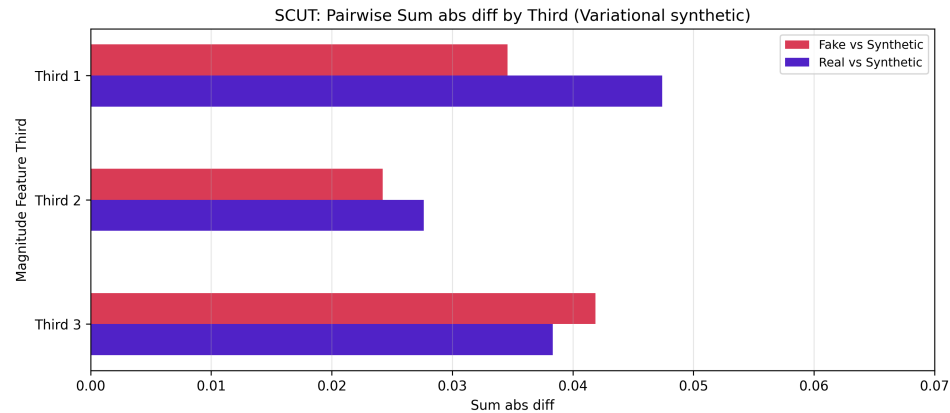


⇒ SCUT: Third 3 is messing up result → *Overall best: 1.*

2026-02-04
Image Processing – Fourier Extraction
└ Experiments Task 2

└ Explanation – SCUT - residual





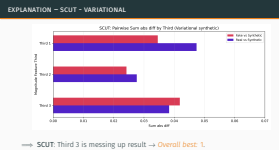
⇒ SCUT: Third 3 is messing up result → *Overall best: 1.*

2026-02-04

Image Processing – Fourier Extraction

└ Experiments Task 2

└ Explanation – SCUT - variational



2026-02-04

Image Processing – Fourier Extraction
└ Results Task 2

Results Task 2

Results Task 2

Top: **Residual** Synthetic Data – Bottom: **Variational** Syntehtic Data

Dataset	Thirds	Step 3 (%)			Step 4 (%)			Step 5 (%)		
		APCER	BPCER	ACER	APCER	BPCER	ACER	APCER	BPCER	ACER
IDIAP	12	0.00	0.83	0.42	0.00	0.42	0.21	0.00	0.42	0.21
PLUS	123	19.12	4.67	11.89	29.28	2.63	15.96	99.44	0.00	49.72
SCUT	1	7.86	0.83	4.35	8.97	0.58	4.78	32.14	0.03	16.08

2026-02-04

Image Processing – Fourier Extraction

└─Results Task 2

└─Results Task 2

RESULTS TASK 2										
Top: Residual Synthetic Data – Bottom: Variational Synthetic Data										
Dataset	Thirds	Step 3 (%)			Step 4 (%)			Step 5 (%)		
		APCER	BPCER	ACER	APCER	BPCER	ACER	APCER	BPCER	ACER
IDIAP	12	0.00	0.83	0.42	0.00	0.42	0.21	0.00	0.42	0.21
PLUS	123	19.12	4.67	11.89	29.28	2.63	15.96	99.44	0.00	49.72
SCUT	1	7.86	0.83	4.35	8.97	0.58	4.78	32.14	0.03	16.08

RESULTS TASK 2

Top: **Residual** Synthetic Data – Bottom: **Variational** Syntehtic Data

Dataset	Thirds	Step 3 (%)			Step 4 (%)			Step 5 (%)		
		APCER	BPCER	ACER	APCER	BPCER	ACER	APCER	BPCER	ACER
IDIAP	12	0.00	0.83	0.42	0.00	0.42	0.21	0.00	0.42	0.21
PLUS	123	19.12	4.67	11.89	29.28	2.63	15.96	99.44	0.00	49.72
SCUT	1	7.86	0.83	4.35	8.97	0.58	4.78	32.14	0.03	16.08

Dataset	Thirds	Step 3 (%)			Step 4 (%)			Step 5 (%)		
		APCER	BPCER	ACER	APCER	BPCER	ACER	APCER	BPCER	ACER
IDIAP	12	0.00	0.83	0.42	0.00	0.42	0.21	0.42	0.42	0.42
PLUS	123	17.57	4.67	11.12	28.28	2.63	15.46	98.44	0.00	49.22
SCUT	1	7.75	1.03	4.39	8.69	0.78	4.74	21.64	0.19	10.92

⇒ **APCER** (false positive) **increases** while **BPCER** (false negative) decreases.

⇒ Once again, **IDIAP** remains unexpectedly **low** over all steps.

2026-02-04

Image Processing – Fourier Extraction

Results Task 2

└ Results Task 2

RESULTS TASK 2									
Top: Residual Synthetic Data - Bottom: Variational Synthetic Data									
Dataset	Thirds	Step 1 (%)			Step 4 (%)			Step 5 (%)	
		APCER	BPCER	ACER	APCER	BPCER	ACER	APCER	ACER
IDIAP	32	0.00	0.83	0.40	0.00	0.42	0.23	0.00	0.42
PLUS	123	19.32	4.67	11.89	29.28	2.83	15.96	99.44	0.00
SCUT	1	7.86	0.83	4.35	8.97	0.58	4.72	32.34	0.03

Dataset	Thirds	Step 1 (%)			Step 4 (%)			Step 5 (%)	
		APCER	BPCER	ACER	APCER	BPCER	ACER	APCER	ACER
IDIAP	32	0.00	0.83	0.40	0.42	0.23	0.42	0.42	0.42
PLUS	123	17.57	4.67	11.12	28.28	2.83	15.48	98.44	0.00
SCUT	1	7.75	1.03	4.29	8.69	0.78	4.74	21.64	0.09

⇒ APCER (false positive) increases while BPCER (false negative) decreases.
 ⇒ Once again, IDIAP remains unexpectedly low over all steps.

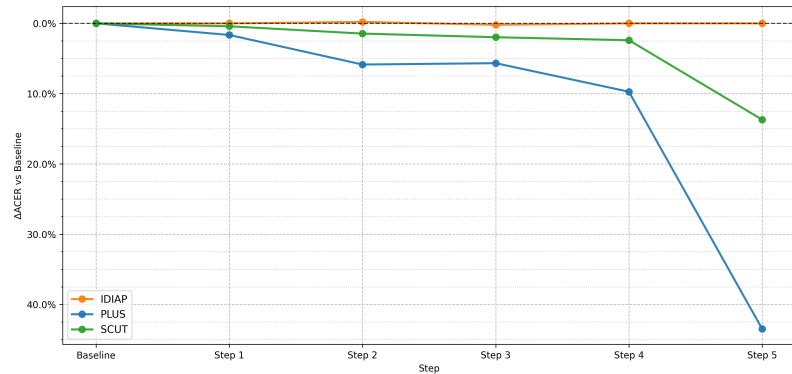
2026-02-04

Image Processing – Fourier Extraction
└─ Final Results

Final Results

Final Results

FINAL RESULTS



QUESTION: Can synthetic spoof data replace real spoof data?

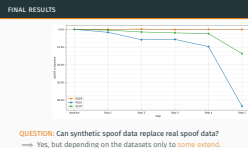
⇒ Yes, but depending on the datasets only to **some extend**.

2026-02-04

Image Processing – Fourier Extraction

└ Final Results

└ Final Results



QUESTION: Can synthetic spoof data replace real spoof data?
⇒ Yes, but depending on the datasets only to **some extend**.

Questions?

2026-02-04

Image Processing – Fourier Extraction
└─ Final Results

Questions?